

Online Resources

Detailed Numerical Experiments

Robust Optimization Under Sparse Uncertainty

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Mathematical Programming

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This supplement reports each run behind the summary displays; objective directions and gap definitions follow the corresponding experiments.

1 Sparse Signal Recovery

Table S1: Detailed sparse signal recovery results.

Instance (n, m, s)	AGP		IHT		PSS	
	Loss	Time (s)	Loss	Time (s)	Loss	Time (s)
#1 (30000,100,5)	0.5948	0.47	1.5639	7.88	0.2444	0.60
#2 (30000,100,5)	1.9787	0.17	4.0152	7.40	1.7916	0.42
#3 (30000,100,5)	0.2293	0.22	4.2776	7.33	1.8002	0.39
#4 (30000,100,5)	0.9656	0.33	3.0069	7.40	1.2840	0.43
#5 (30000,100,5)	1.5095	0.20	4.7572	7.50	2.2049	0.41
#6 (30000,100,10)	0.9182	0.33	4.9576	7.37	1.7959	1.14
#7 (30000,100,10)	1.1254	0.32	5.2040	7.57	1.7475	0.92
#8 (30000,100,10)	0.6663	0.36	10.0229	7.64	1.5928	1.00
#9 (30000,100,10)	0.7860	0.30	4.1987	7.53	0.6592	1.62
#10 (30000,100,10)	1.0733	0.29	4.6472	7.25	1.8403	1.42
#11 (30000,100,20)	1.4281	0.45	8.1492	7.34	2.2593	5.37
#12 (30000,100,20)	0.9696	0.24	8.1493	7.05	1.3207	4.52
#13 (30000,100,20)	0.7585	0.39	5.3076	6.90	0.9456	4.59
#14 (30000,100,20)	0.3322	0.28	3.5001	7.06	0.5627	4.32
#15 (30000,100,20)	0.6809	0.41	4.0033	7.04	1.8472	4.27
#16 (50000,100,5)	0	0.59	2.9128	12.93	0	1.20
#17 (50000,100,5)	3.1723	0.60	9.7422	13.49	0	0.93
#18 (50000,100,5)	0.6141	0.44	3.3429	13.78	1.1163	1.16
#19 (50000,100,5)	0.0514	0.59	2.5970	13.73	0.8185	1.11
#20 (50000,100,5)	1.2275	0.44	3.9558	13.50	1.0982	0.79
#21 (50000,100,10)	0.3876	0.70	5.4457	13.71	0.9353	2.94
#22 (50000,100,10)	0.6729	0.66	5.0327	13.39	0.7710	2.08
#23 (50000,100,10)	0.8712	0.65	8.0828	13.47	1.3397	1.49
#24 (50000,100,10)	1.1382	0.54	9.1042	13.39	1.3831	2.74
#25 (50000,100,10)	2.9287	0.58	11.7412	13.54	1.7181	2.51
#26 (50000,100,20)	0.8364	0.80	21.2350	13.29	1.7404	6.96

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Table S1: Sparse signal recovery (continued).

Instance (n, m, s)	AGP		IHT		PSS	
	Loss	Time (s)	Loss	Time (s)	Loss	Time (s)
#27 (50000,100,20)	0.3878	0.68	8.5209	12.59	0.5509	6.54
#28 (50000,100,20)	0.8358	1.30	7.1836	12.43	1.2302	7.54
#29 (50000,100,20)	0.4033	0.59	11.0566	12.88	0.8892	8.53
#30 (50000,100,20)	1.6134	0.94	7.5934	12.44	1.2460	7.33
#31 (30000,200,5)	0	0.34	0.5219	11.88	0	0.67
#32 (30000,200,5)	0	0.29	1.2144	12.55	0	0.75
#33 (30000,200,5)	0	0.38	2.8769	12.04	0	0.73
#34 (30000,200,5)	0	0.29	2.6324	12.60	0	0.88
#35 (30000,200,5)	0	0.33	2.3946	11.78	2.1587	0.45
#36 (30000,200,10)	0.8802	0.67	6.4024	12.84	1.6985	2.03
#37 (30000,200,10)	0.2872	0.44	1.7014	13.08	0.2043	4.25
#38 (30000,200,10)	1.4350	0.34	2.9027	12.06	1.2225	1.64
#39 (30000,200,10)	0	0.54	1.8687	12.45	0.4080	2.81
#40 (30000,200,10)	0	0.36	2.9824	12.08	1.5494	1.55
#41 (30000,200,20)	2.0881	0.72	6.6531	12.19	3.1459	4.77
#42 (30000,200,20)	1.3094	0.59	5.4618	11.94	2.6801	3.83
#43 (30000,200,20)	1.0016	0.56	4.3891	12.08	1.7395	6.65
#44 (30000,200,20)	1.1459	0.50	3.1816	12.45	1.8090	4.13
#45 (30000,200,20)	0.7907	0.96	3.7111	12.20	1.6622	4.94
#46 (50000,200,5)	0	0.93	5.6571	23.13	0	1.12
#47 (50000,200,5)	0	0.84	1.3331	23.52	0	1.19
#48 (50000,200,5)	0.9349	0.81	1.8573	23.43	0	1.26
#49 (50000,200,5)	0	0.84	6.9092	23.37	0	1.06
#50 (50000,200,5)	0	1.03	3.6455	23.57	1.3511	1.25
#51 (50000,200,10)	1.4390	1.29	6.3916	23.43	1.7590	2.77
#52 (50000,200,10)	0.4809	1.59	5.3121	23.31	0.6220	4.57
#53 (50000,200,10)	0	1.30	11.6907	23.22	1.8477	3.47
#54 (50000,200,10)	0	1.05	4.5230	23.49	0.3142	4.42
#55 (50000,200,10)	0	1.22	2.5388	23.29	0.3548	4.50
#56 (50000,200,20)	1.9559	1.15	4.2751	23.14	2.6981	11.01
#57 (50000,200,20)	1.6358	1.68	7.8231	23.29	3.6864	5.96
#58 (50000,200,20)	2.0973	1.67	11.9856	23.75	3.9059	7.94
#59 (50000,200,20)	1.1412	1.37	6.4544	23.81	2.5841	8.56
#60 (50000,200,20)	0.2475	1.48	2.4640	23.67	1.6092	13.51

2 Sparse Logistic Regression

Table S2: Detailed sparse logistic regression results on LIBSVM data sets.

Instance (data, m, n, s)	AGP		IHT		PSS	
	Loss	Time (s)	Loss	Time (s)	Loss	Time (s)
#1 (a5a, 6414, 123, 5)	0.3758	0.20	0.3841	0.95	0.3892	0.23
#2 (a5a, 6414, 123, 5)	0.3758	0.07	0.3841	0.88	0.3719	0.43

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Table S2: Sparse logistic regression (continued).

Instance (data, m , n , s)	AGP		IHT		PSS	
	Loss	Time (s)	Loss	Time (s)	Loss	Time (s)
#3 (a5a, 6414, 123, 5)	0.3941	0.13	0.5038	0.87	0.3806	0.60
#4 (a5a, 6414, 123, 5)	0.3758	0.07	0.4333	0.86	0.3719	0.40
#5 (a5a, 6414, 123, 5)	0.3806	0.12	0.5206	0.86	0.3719	0.68
#6 (a5a, 6414, 123, 10)	0.3510	1.50	0.3758	0.78	0.3490	0.80
#7 (a5a, 6414, 123, 10)	0.3559	0.66	0.3673	0.89	0.3607	1.18
#8 (a5a, 6414, 123, 10)	0.3522	1.39	0.3587	0.89	0.3632	1.44
#9 (a5a, 6414, 123, 10)	0.3489	1.41	0.3725	0.91	0.3552	1.39
#10 (a5a, 6414, 123, 10)	0.3550	0.56	0.3725	0.90	0.3513	1.07
#11 (a5a, 6414, 123, 20)	0.3358	1.51	0.3547	0.83	0.3409	2.65
#12 (a5a, 6414, 123, 20)	0.3366	0.43	0.3439	0.93	0.3445	3.55
#13 (a5a, 6414, 123, 20)	0.3375	1.52	0.3545	0.84	0.3414	2.74
#14 (a5a, 6414, 123, 20)	0.3361	1.47	0.3584	0.91	0.3400	3.39
#15 (a5a, 6414, 123, 20)	0.3397	0.65	0.3585	0.83	0.3406	3.26
#16 (a6a, 11220, 123, 5)	0.3753	0.19	0.3826	1.64	0.3897	0.50
#17 (a6a, 11220, 123, 5)	0.3753	0.20	0.3826	1.71	0.3714	0.83
#18 (a6a, 11220, 123, 5)	0.3924	0.30	0.5002	1.63	0.3786	1.15
#19 (a6a, 11220, 123, 5)	0.3753	0.17	0.5124	1.62	0.3714	0.88
#20 (a6a, 11220, 123, 5)	0.3786	0.18	0.4321	1.64	0.3714	1.31
#21 (a6a, 11220, 123, 10)	0.3499	1.46	0.3826	1.67	0.3497	1.96
#22 (a6a, 11220, 123, 10)	0.3495	0.53	0.3650	1.69	0.3666	1.89
#23 (a6a, 11220, 123, 10)	0.3485	1.00	0.3607	1.71	0.3558	2.18
#24 (a6a, 11220, 123, 10)	0.3515	0.35	0.3786	1.74	0.3587	2.65
#25 (a6a, 11220, 123, 10)	0.3555	0.92	0.3786	1.72	0.3521	2.07
#26 (a6a, 11220, 123, 20)	0.3371	1.89	0.3524	1.74	0.3418	5.01
#27 (a6a, 11220, 123, 20)	0.3371	1.41	0.3445	1.65	0.3475	9.83
#28 (a6a, 11220, 123, 20)	0.3364	1.81	0.3534	1.74	0.3399	5.91
#29 (a6a, 11220, 123, 20)	0.3366	3.01	0.3617	1.75	0.3407	5.93
#30 (a6a, 11220, 123, 20)	0.3419	1.78	0.3600	1.74	0.3406	43.42
#31 (a7a, 16100, 123, 5)	0.3787	0.26	0.3839	2.50	0.3726	1.79
#32 (a7a, 16100, 123, 5)	0.3787	0.41	0.3853	2.64	0.3726	1.25
#33 (a7a, 16100, 123, 5)	0.3918	0.49	0.5043	2.45	0.3726	1.36
#34 (a7a, 16100, 123, 5)	0.3799	0.48	0.5140	2.47	0.3726	1.19
#35 (a7a, 16100, 123, 5)	0.3799	0.46	0.5176	2.46	0.3726	1.85
#36 (a7a, 16100, 123, 10)	0.3547	2.03	0.3839	2.53	0.3565	2.30
#37 (a7a, 16100, 123, 10)	0.3493	1.07	0.3729	2.63	0.3666	3.58
#38 (a7a, 16100, 123, 10)	0.3582	2.55	0.3761	2.63	0.3655	3.75
#39 (a7a, 16100, 123, 10)	0.3547	2.41	0.3796	2.70	0.3606	3.92
#40 (a7a, 16100, 123, 10)	0.3491	1.30	0.3797	2.68	0.3536	3.29
#41 (a7a, 16100, 123, 20)	0.3396	2.89	0.3598	2.66	0.3441	6.34
#42 (a7a, 16100, 123, 20)	0.3385	1.80	0.3492	2.75	0.3431	28.21
#43 (a7a, 16100, 123, 20)	0.3374	2.72	0.3605	2.58	0.3447	7.55
#44 (a7a, 16100, 123, 20)	0.3371	2.66	0.3608	2.63	0.3419	8.59
#45 (a7a, 16100, 123, 20)	0.3408	1.35	0.3566	2.59	0.3417	26.33
#46 (a8a, 22696, 123, 5)	0.3782	0.46	0.4057	3.55	0.3722	2.52
#47 (a8a, 22696, 123, 5)	0.3782	0.52	0.4048	3.69	0.3722	1.81
#48 (a8a, 22696, 123, 5)	0.3909	0.71	0.5028	3.54	0.3722	1.92

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Table S2: Sparse logistic regression (continued).

Instance (data, m , n , s)	AGP		IHT		PSS	
	Loss	Time (s)	Loss	Time (s)	Loss	Time (s)
#49 (a8a, 22696, 123, 5)	0.3782	0.68	0.5101	3.50	0.3722	1.79
#50 (a8a, 22696, 123, 5)	0.3918	0.37	0.5148	3.54	0.3722	2.55
#51 (a8a, 22696, 123, 10)	0.3539	3.98	0.3912	3.54	0.3561	3.10
#52 (a8a, 22696, 123, 10)	0.3484	0.57	0.3727	3.72	0.3662	4.80
#53 (a8a, 22696, 123, 10)	0.3554	3.31	0.3755	3.77	0.3601	5.49
#54 (a8a, 22696, 123, 10)	0.3591	5.65	0.3791	3.96	0.3655	5.58
#55 (a8a, 22696, 123, 10)	0.3592	3.56	0.3791	3.94	0.3535	4.55
#56 (a8a, 22696, 123, 20)	0.3405	6.81	0.3590	4.02	0.3461	11.37
#57 (a8a, 22696, 123, 20)	0.3368	2.64	0.3460	4.03	0.3455	11.17
#58 (a8a, 22696, 123, 20)	0.3373	6.75	0.3697	4.02	0.3437	12.90
#59 (a8a, 22696, 123, 20)	0.3373	5.82	0.3535	4.00	0.3447	17.65
#60 (a8a, 22696, 123, 20)	0.3387	2.86	0.3566	4.08	0.3432	88.31

3 Portfolio Optimization under Sparse Stress Profiles

Table S3: Detailed portfolio results for the 49 Fama-French industry portfolios.

Tail	(q , s)	Quantity	Exact	AGP	IHT
6.7%	(24, 3)	Status	Optimal		
		Objective	-0.608601	-0.680045	-0.867723
		Time (s)	13.347	0.444	0.117
		Iterations	125	17	27
		Cuts	124	148	193
		Shortfall (%)		13.33	33.13
		Evaluation time (s)		0.0004	0.0004
10%	(36, 3)	Status	Optimal		
		Objective	-0.682606	-0.741690	-0.817179
		Time (s)	65.761	0.297	0.209
		Iterations	172	35	35
		Cuts	171	251	227
		Shortfall (%)		10.32	20.52
		Evaluation time (s)		0.0014	0.0014
15%	(54, 3)	Status	Optimal		
		Objective	-0.730607	-0.808257	-1.289916
		Time (s)	527.869	0.303	0.218
		Iterations	237	25	32
		Cuts	236	190	210
		Shortfall (%)		15.22	52.58

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Table S3: Detailed portfolio results (continued).

Tail	(q, s)	Quantity	Exact	AGP	IHT
		Evaluation time (s)		0.0045	0.0047
25%	(90, 3)	Status	Time limit		
		Objective	n/a	-0.953634	-0.982332
		Time (s)	1000.007	0.635	0.575
		Iterations	56	34	35
		Cuts	55	300	333
		Shortfall (%)		21.03	29.12
		Evaluation time (s)		0.0307	0.0307
30%	(108, 3)	Status	Time limit		
		Objective	n/a	-1.078418	-1.380892
		Time (s)	1000.000	0.930	0.261
		Iterations	n/a	28	22
		Cuts	n/a	235	183
		Shortfall (%)		31.39	55.70
		Evaluation time (s)		0.0437	0.0450

4 Linear Optimization with Implementation Error

In Table S4, Bound is the lower bound returned by RL, and Gap is its percentage difference from the exact EL optimum. Exact violation is the largest excess over a robust constraint, computed by binary separation at the returned decision.

Table S4: Detailed implementation error results on NETLIB test instances.

Instance	(n, m, L)	s	Quantity	Exact (EL)	RL lower bound
SCFXM1	(457, 143, 240)	10	Value	18591.9499	18589.4484
			Time (s)	4.997	0.856
			Iterations	4	5
			Cuts	85	87
			Exact violation	0.00	6.16228
			Gap (%)		0.01346
		20	Value	18597.1073	18597.1073
			Time (s)	2.935	1.363
			Iterations	7	8
			Cuts	112	115
			Exact violation	0.00	0.00
			Gap (%)		0.00

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Table S4: Detailed implementation error results (continued).

Instance	(n, m, L)	s	Quantity	Exact (EL)	RL lower bound
SCFXM3	(1371, 429, 480)	10	Value	55563.4905	55553.5742
			Time (s)	13.171	4.201
			Iterations	6	5
			Cuts	280	271
			Exact violation	0.00	4.96634
			Gap (%)		0.01785
		20	Value	55658.6671	55646.4813
			Time (s)	34.924	6.677
			Iterations	12	8
			Cuts	361	307
			Exact violation	0.00	10.68175
			Gap (%)		0.02189
25FV47	(1571, 305, 600)	10	Value	5601.5376	5600.9651
			Time (s)	26.857	8.651
			Iterations	13	11
			Cuts	235	217
			Exact violation	0.00	0.69740
			Gap (%)		0.01022
		20	Value	5615.4199	5613.9875
			Time (s)	58.621	13.468
			Iterations	14	16
			Cuts	317	279
			Exact violation	0.00	1.00643
			Gap (%)		0.02551

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Table S4: Detailed implementation error results (continued).

Instance	(n, m, L)	s	Quantity	Exact (EL)	RL lower bound
WOODW	(8405, 13, 1200)	10	Value	1.3148	1.3147
			Time (s)	2.166	0.861
			Iterations	11	9
			Cuts	12	10
			Exact violation	0.00	0.00008
			Gap (%)		0.00684
		20	Value	1.3189	1.3188
			Time (s)	2.719	1.186
			Iterations	11	11
			Cuts	12	12
			Exact violation	0.00	0.00007
			Gap (%)		0.00633
FIT2D	(10500, 24, 1500)	10	Value	-68449.1238	-68449.3366
			Time (s)	9.693	7.268
			Iterations	2	2
			Cuts	19	19
			Exact violation	0.00	0.88072
			Gap (%)		0.00031
		20	Value	-68438.4040	-68438.9662
			Time (s)	17.885	9.356
			Iterations	3	3
			Cuts	24	22
			Exact violation	0.00	3.71566
			Gap (%)		0.00082

5 Wireless Downlink Planning under Sparse Jamming

Table S5: Detailed static wireless results from the DICHASUS measurements.

Instance	Tuple	Quantity	SP	RL	EL
1.1	(47, 200, 20, 8, 67)	Objective	0.02491	0.03710	0.03776
		Time (s)	0.40	0.27	8.91
		Iterations	5	5	4
		Cuts	8	9	9
		Exact violation	0.00667	0.00034	0.00
1.2	(47, 200, 20, 8, 67)	Objective	0.02506	0.03803	0.03803
		Time (s)	0.24	0.21	10.27

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Table S5: Detailed static wireless results (continued).

Instance	Tuple	Quantity	SP	RL	EL
		Iterations	4	4	5
		Cuts	7	9	11
		Exact violation	0.00589	0.00	0.00
1.3	(47, 200, 20, 8, 67)	Objective	0.02448	0.03624	0.03655
		Time (s)	0.23	0.22	8.77
		Iterations	4	4	4
		Cuts	8	8	7
		Exact violation	0.00636	0.00015	0.00
2.1	(47, 250, 30, 12, 84)	Objective	0.02387	0.03102	0.03201
		Time (s)	0.46	0.48	28.75
		Iterations	4	5	6
		Cuts	7	12	13
		Exact violation	0.00445	0.00084	0.00
2.2	(47, 250, 30, 12, 84)	Objective	0.02443	0.03472	0.03511
		Time (s)	0.72	0.39	19.97
		Iterations	5	4	4
		Cuts	9	9	9
		Exact violation	0.00577	0.00026	0.00
2.3	(47, 250, 30, 12, 84)	Objective	0.02558	0.03667	0.03674
		Time (s)	0.56	0.29	21.03
		Iterations	5	3	4
		Cuts	9	6	8
		Exact violation	0.00554	0.00004	0.00

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Table S5: Detailed static wireless results (continued).

Instance	Tuple	Quantity	SP	RL	EL
3.1	(47, 300, 40, 14, 100)	Objective	0.02490	0.03374	0.03439
		Time (s)	0.72	0.78	55.68
		Iterations	4	5	7
		Cuts	7	11	16
		Exact violation	0.00524	0.00043	0.00
3.2	(47, 300, 40, 14, 100)	Objective	0.02705	0.03468	0.03493
		Time (s)	0.96	0.62	35.05
		Iterations	4	4	4
		Cuts	7	8	8
		Exact violation	0.00378	0.00021	0.00
3.3	(47, 300, 40, 14, 100)	Objective	0.02440	0.03292	0.03333
		Time (s)	0.75	0.72	34.72
		Iterations	4	4	4
		Cuts	7	9	8
		Exact violation	0.00544	0.00042	0.00
4.1	(47, 300, 50, 14, 100)	Objective	0.02521	0.03312	0.03355
		Time (s)	1.63	1.17	64.53
		Iterations	7	6	6
		Cuts	12	14	15
		Exact violation	0.00448	0.00030	0.00
4.2	(47, 300, 50, 14, 100)	Objective	0.03023	0.03865	0.03943
		Time (s)	0.46	0.77	57.42
		Iterations	2	4	5
		Cuts	3	9	10
		Exact violation	0.00392	0.00052	0.00
4.3	(47, 300, 50, 14, 100)	Objective	0.02547	0.03326	0.03386
		Time (s)	1.16	0.77	55.51
		Iterations	5	4	5
		Cuts	10	9	10
		Exact violation	0.00531	0.00038	0.00

6 Adjustable Wireless Planning with Shared Recovery

Table S6: Detailed adjustable wireless lower bounds from DICHASUS data.

Instance	Tuple	Quantity	DB-SP	DB-RL	DB-EL
1.1	(47, 25, 250, 12, 84)	Bound	0.02355	0.02887	0.02943
		Time (s)	6.28	1.24	31.25
		Iterations	14	8	10
		Cuts	14	8	10
		Final violation	0.00	0.00	0.00
1.2	(47, 25, 250, 12, 84)	Bound	0.02408	0.02808	0.02815
		Time (s)	5.56	1.76	39.12
		Iterations	6	12	10
		Cuts	6	12	10
		Final violation	0.00	0.00	0.00
1.3	(47, 25, 250, 12, 84)	Bound	0.02445	0.02904	0.02936
		Time (s)	4.34	1.54	29.97
		Iterations	7	10	9
		Cuts	7	10	9
		Final violation	0.00	0.00	0.00
2.1	(47, 30, 300, 14, 100)	Bound	0.02386	0.02904	0.02912
		Time (s)	8.17	2.97	65.17
		Iterations	13	14	15
		Cuts	13	14	15
		Final violation	0.00	0.00	0.00
2.2	(47, 30, 300, 14, 100)	Bound	0.02420	0.02985	0.03221
		Time (s)	3.06	1.29	49.26
		Iterations	7	7	10
		Cuts	7	7	10
		Final violation	0.00	0.00	0.00

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Table S6: Detailed adjustable wireless lower bounds (continued).

Instance	Tuple	Quantity	DB-SP	DB-RL	DB-EL
2.3	(47, 30, 300, 14, 100)	Bound	0.02421	0.03214	0.03299
		Time (s)	7.18	2.01	54.75
		Iterations	13	10	13
		Cuts	13	10	13
		Final violation	0.00	0.00	0.00
3.1	(47, 35, 300, 14, 100)	Bound	0.02397	0.03083	0.03103
		Time (s)	13.23	2.75	56.42
		Iterations	16	13	13
		Cuts	16	13	13
		Final violation	0.00	0.00	0.00
3.2	(47, 35, 300, 14, 100)	Bound	0.02492	0.03131	0.03205
		Time (s)	2.93	1.80	46.85
		Iterations	7	9	13
		Cuts	7	9	13
		Final violation	0.00	0.00	0.00
3.3	(47, 35, 300, 14, 100)	Bound	0.02465	0.03193	0.03226
		Time (s)	3.52	1.67	41.93
		Iterations	8	8	8
		Cuts	8	8	8
		Final violation	0.00	0.00	0.00

Table S7: Small-instance adjustable wireless results with an exact reference.

Instance	Quantity	Exact Full	DB-SP	DB-RL	DB-EL	AADR Exact	RO-EL
1	Objective	0.00927	0.00927	0.00927	0.00927	0.00927	0.01208
	Time (s)	0.33	0.91	0.62	2.03	0.03	0.00
	Iterations	1	4	4	4	1	1
	Cuts	0	4	4	4	0	0
2	Objective	0.00755	0.00755	0.00755	0.00755	0.00755	0.00949
	Time (s)	0.01	0.66	0.47	1.39	0.01	0.00
	Iterations	1	4	4	4	1	1
	Cuts	0	4	4	4	0	0
3	Objective	0.00861	0.00861	0.00861	0.00861	0.00883	0.01137
	Time (s)	0.01	1.14	1.21	2.63	0.01	0.00
	Iterations	1	9	9	9	1	1
	Cuts	0	9	9	9	0	0

Table S8: Affine and static policies for the adjustable DICHASUS instances.

Instance	Tuple	Quantity	AADR-RL	RO-EL
1.1	(47, 25, 250, 12, 84)	Value	0.03050	0.03072
		Time (s)	5.41	13.64
		Iterations		2
		Cuts		25
1.2	(47, 25, 250, 12, 84)	Value	0.02953	0.02989
		Time (s)	4.96	12.72
		Iterations		2
		Cuts		21
1.3	(47, 25, 250, 12, 84)	Value	0.02997	0.03075
		Time (s)	7.32	12.57
		Iterations		2
		Cuts		27
2.1	(47, 30, 300, 14, 100)	Value	0.03050	0.03105
		Time (s)	12.75	20.23
		Iterations		2
		Cuts		29
2.2	(47, 30, 300, 14, 100)	Value	0.03340	0.03407
		Time (s)	11.21	18.98
		Iterations		2
		Cuts		30
2.3	(47, 30, 300, 14, 100)	Value	0.03392	0.03542
		Time (s)	10.76	18.42
		Iterations		2
		Cuts		28
3.1	(47, 35, 300, 14, 100)	Value	0.03192	0.03247
		Time (s)	11.14	21.15
		Iterations		2
		Cuts		35
3.2	(47, 35, 300, 14, 100)	Value	0.03457	0.03529
		Time (s)	12.12	23.48
		Iterations		2
		Cuts		37
3.3	(47, 35, 300, 14, 100)	Value	0.03312	0.03414
		Time (s)	13.18	23.19
		Iterations		2
		Cuts		32